

Circutor CVM-C11

Modbus RTU Register Map

Product code M58541 | CVM-C11-ITF-IN-485-ICT2

Protocol basics

Mode: Modbus RTU. **Read functions:** 0x03 and 0x04 (integer registers). **Write functions:** 0x05 (single coil), 0x10 (multiple registers).

Every measurement value is a 32-bit integer spanning 2 consecutive registers. All addresses are hexadecimal. Maximum 20 variables (40 registers) per frame. Apply the scale in the Unit column to obtain engineering units.

Addressing note: these are protocol addresses (base 0). If your master uses PLC-style 3xxxx / 4xxxx addressing, add the offset and +1 yourself.

Instantaneous / Max / Min measurements (Table 1)

Parameter	Inst (hex)	Max (hex)	Min (hex)	Unit / scale
Voltage L1	00-01	106-107	164-165	V x10
Current L1	02-03	108-109	166-167	mA
Active power L1	04-05	10A-10B	168-169	W
Inductive reactive power L1	06-07	10C-10D	16A-16B	var
Capacitive reactive power L1	08-09	10E-10F	16C-16D	var
Apparent power L1	0A-0B	110-111	16E-16F	VA
Power factor L1	0C-0D	112-113	170-171	x100
Cos ϕ L1	0E-0F	114-115	172-173	x100
Voltage L2	10-11	116-117	174-175	V x10
Current L2	12-13	118-119	176-177	mA
Active power L2	14-15	11A-11B	178-179	W
Inductive reactive power L2	16-17	11C-11D	17A-17B	var
Capacitive reactive power L2	18-19	11E-11F	17C-17D	var
Apparent power L2	1A-1B	120-121	17E-17F	VA
Power factor L2	1C-1D	122-123	180-181	x100
Cos ϕ L2	1E-1F	124-125	182-183	x100
Voltage L3	20-21	126-127	184-185	V x10
Current L3	22-23	128-129	186-187	mA
Active power L3	24-25	12A-12B	188-189	W
Inductive reactive power L3	26-27	12C-12D	18A-18B	var
Capacitive reactive power L3	28-29	12E-12F	18C-18D	var
Apparent power L3	2A-2B	130-131	18E-18F	VA
Power factor L3	2C-2D	132-133	190-191	x100
Cos ϕ L3	2E-2F	134-135	192-193	x100
Active power III (3-phase)	30-31	136-137	194-195	W
Inductive reactive power III	32-33	138-139	196-197	var
Capacitive reactive power III	34-35	13A-13B	198-199	var
Apparent power III	36-37	13C-13D	19A-19B	VA

Parameter	Inst (hex)	Max (hex)	Min (hex)	Unit / scale
Power factor III	38-39	13E-13F	19C-19D	x100
Cos ϕ III	3A-3B	140-141	19E-19F	x100
Frequency (L1)	3C-3D	142-143	1A0-1A1	Hz x100
Voltage L1-L2	3E-3F	144-145	1A2-1A3	V x10
Voltage L2-L3	40-41	146-147	1A4-1A5	V x10
Voltage L3-L1	42-43	148-149	1A6-1A7	V x10
Neutral current	44-45	14A-14B	1A8-1A9	mA
%THD voltage L1	46-47	14C-14D	1AA-1AB	% x10
%THD voltage L2	48-49	14E-14F	1AC-1AD	% x10
%THD voltage L3	4A-4B	150-151	1AE-1AF	% x10
%THD current L1	4C-4D	152-153	1B0-1B1	% x10
%THD current L2	4E-4F	154-155	1B2-1B3	% x10
%THD current L3	50-51	156-157	1B4-1B5	% x10
Max demand, active power III	52-53	158-159	n/a	W
Max demand, apparent power III	54-55	15A-15B	n/a	VA
Max demand, current AVG	56-57	15C-15D	n/a	mA
Max demand, current L1	58-59	15E-15F	n/a	mA
Max demand, current L2	5A-5B	160-161	n/a	mA
Max demand, current L3	5C-5D	162-163	n/a	mA
Max demand, ind. reactive power III	200-201	204-205	n/a	kvarL
Max demand, cap. reactive power III	202-203	206-207	n/a	kvarC

Energy and tariff registers (Table 2)

Parameter	T1	T2	T3	Total	Unit
Active energy consumed (kW range)	5E-5F	88-89	B2-B3	DC-DD	kWh
Active energy consumed (W range)	60-61	8A-8B	B4-B5	DE-DF	Wh
Ind. reactive consumed (kvarh)	62-63	8C-8D	B6-B7	E0-E1	kvarh
Ind. reactive consumed (varh)	64-65	8E-8F	B8-B9	E2-E3	varh
Cap. reactive consumed (kvarh)	66-67	90-91	BA-BB	E4-E5	kvarh
Cap. reactive consumed (varh)	68-69	92-93	BC-BD	E6-E7	varh
Apparent consumed (kVAh)	6A-6B	94-95	BE-BF	E8-E9	kVAh
Apparent consumed (VAh)	6C-6D	96-97	C0-C1	EA-EB	VAh
CO2 emissions consumed	6E-6F	98-99	C2-C3	EC-ED	x10
Cost consumed	70-71	9A-9B	C4-C5	EE-EF	x10
Active energy generated (kW range)	72-73	9C-9D	C6-C7	F0-F1	kWh
Active energy generated (W range)	74-75	9E-9F	C8-C9	F2-F3	Wh
Ind. reactive generated (kvarh)	76-77	A0-A1	CA-CB	F4-F5	kvarh
Ind. reactive generated (varh)	78-79	A2-A3	CC-CD	F6-F7	varh
Cap. reactive generated (kvarh)	7A-7B	A4-A5	CE-CF	F8-F9	kvarh
Cap. reactive generated (varh)	7C-7D	A6-A7	D0-D1	FA-FB	varh
Apparent generated (kVAh)	7E-7F	A8-A9	D2-D3	FC-FD	kVAh
Apparent generated (VAh)	80-81	AA-AB	D4-D5	FE-EF	VAh

Parameter	T1	T2	T3	Total	Unit
CO2 emissions generated	82-83	AC-AD	D6-D7	100-101	x10
Cost generated	84-85	AE-AF	D8-D9	102-103	x10
Hours per tariff	86-87	B0-B1	DA-DB	104-105	seconds

Harmonics (Tables 3 and 4)

Voltage harmonics. Fundamental at A28-A29 (L1), A48-A49 (L2), A68-A69 (L3), in V x10. Orders 2 through 31 follow as single registers immediately after each fundamental: L1 A2A (2nd) through A47 (31st); L2 block starts at A4A; L3 block starts at A6A. All harmonic orders in % x10.

Current harmonics. Fundamental at A88-A89 (L1), AA8-AA9 (L2), AC8-AC9 (L3), in mA. Orders 2 through 31 follow as single registers: L1 A8A (2nd) through AA7 (31st); L2 block starts at AAA; L3 block starts at ACA. All harmonic orders in % x10.

Write / reset commands

Action (function 0x05, write FF00)	Address (hex)
Clear energies	834
Clear max/min	838
Reset max demand	839
Clear hour counters (all tariffs)	83D
Clear max-demand peak	83F
Clear energies + max demand + max/min	848
Power status / quadrant (read, function 0x04)	7D1

Source: Circutor CVM-C11 instruction manual, document M361B01, Section 7/8 (Modbus Commands). Cross-check against the current manual at docs.circutor.com before commissioning, as Circutor may revise addresses across firmware versions.